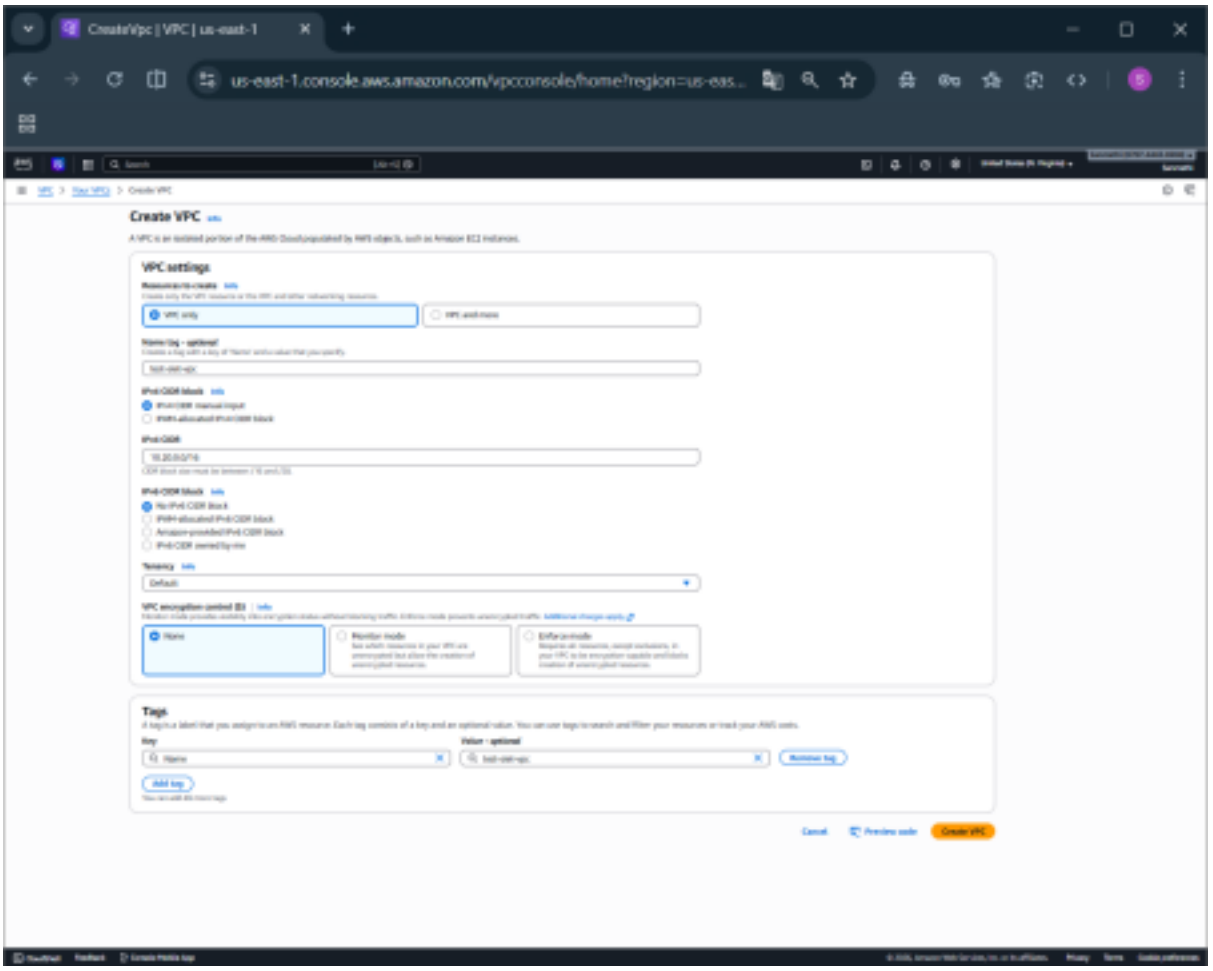
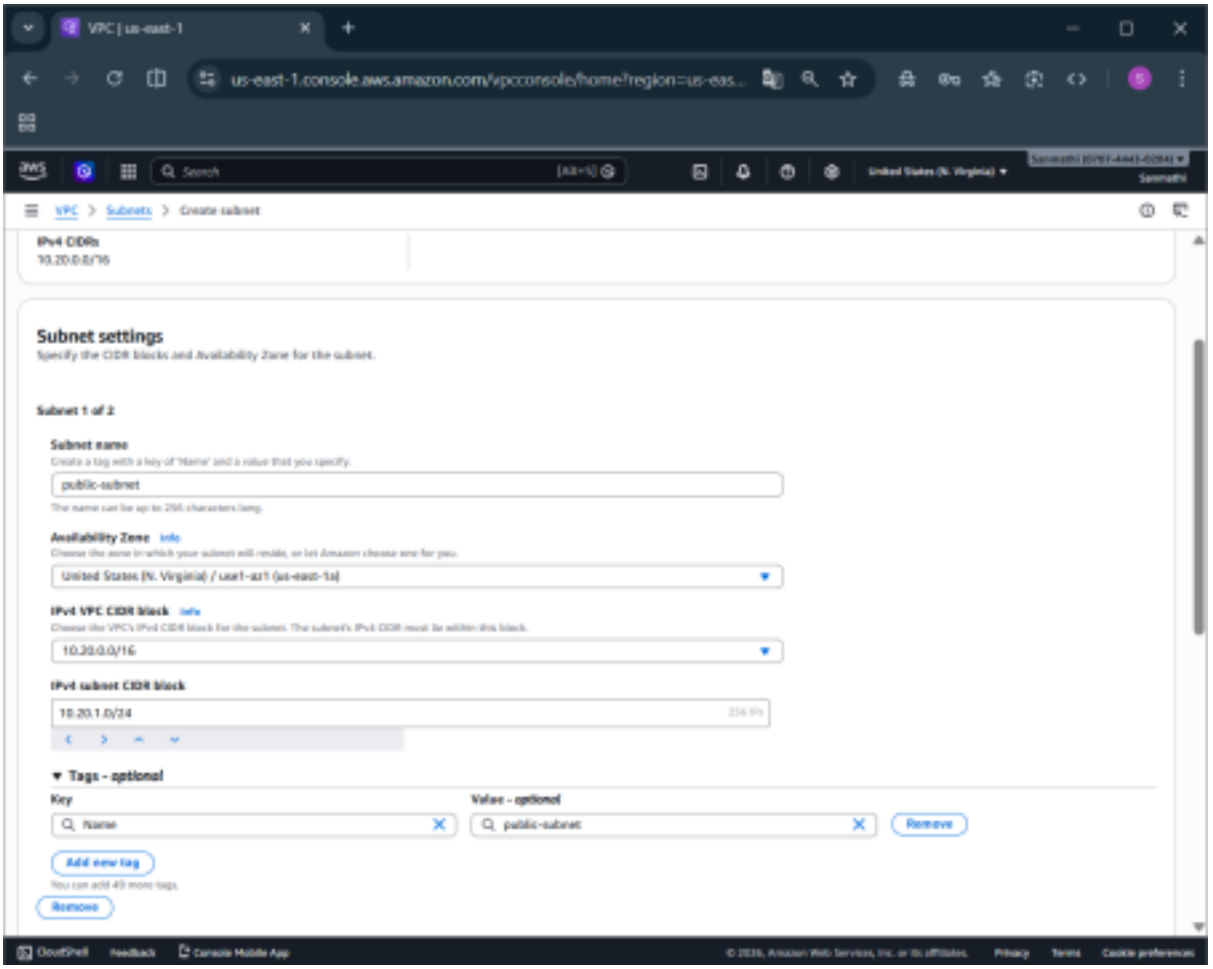
DEVOPS - 18-03-2026

TERRAFORM CONFIGURATION:




VPC | us-east-1
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-eas...
Search
United States (N. Virginia)
Sanvathi
VPC > Subnets > Create subnet
IPv4 CIDRs
10.20.0.0/16
Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.
Subnet 1 of 2
Subnet name
Create a tag with a key of 'Name' and a value that you specify.
public-subnet
The name can be up to 256 characters long.
Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / use1-az1 (us-east-1a)
IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.20.0.0/16
IPv4 subnet CIDR block
10.20.1.0/24
256 IPs
Tags - optional
Key
Value - optional
Name
public-subnet
Remove
Add new tag
You can add 49 more tags.
Remove
CloudShell
Feedback
Console Mobile App
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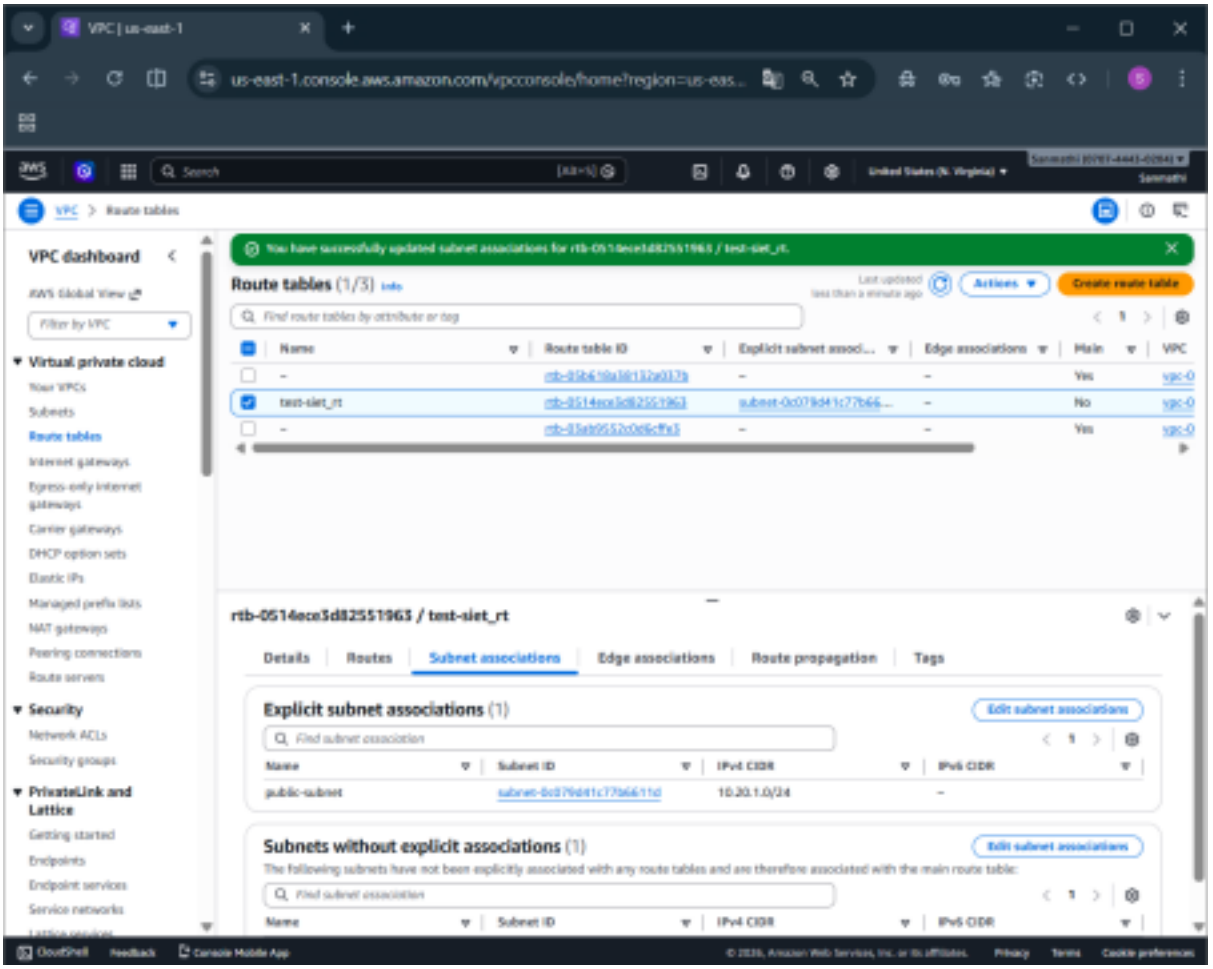
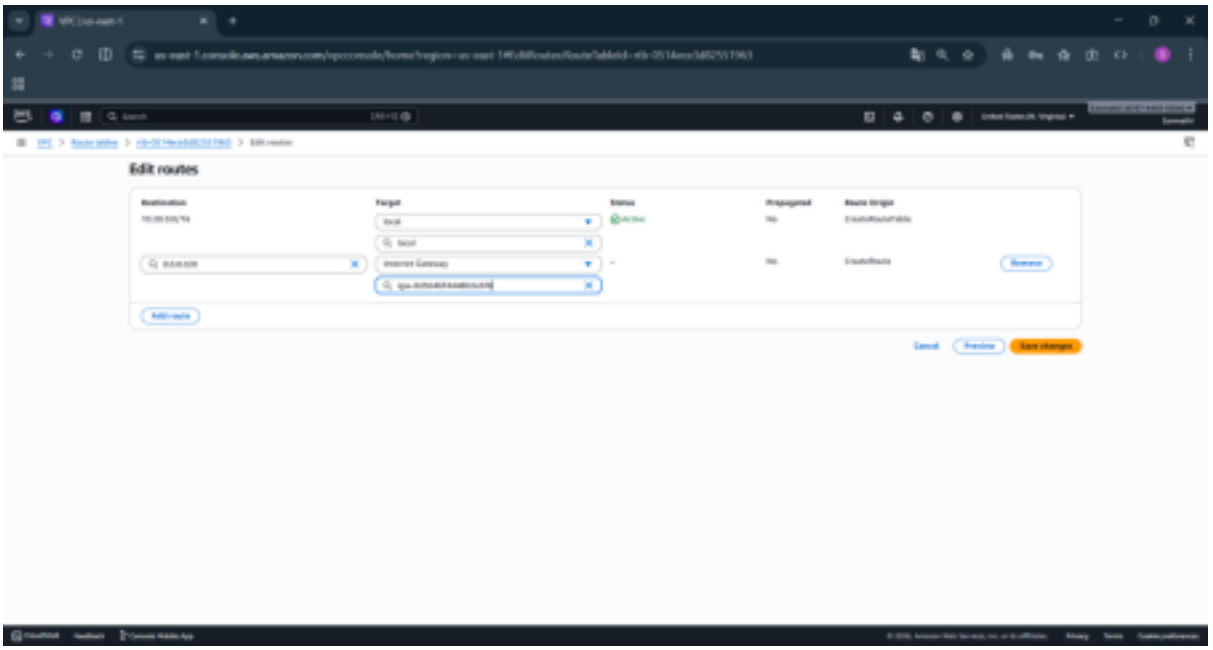
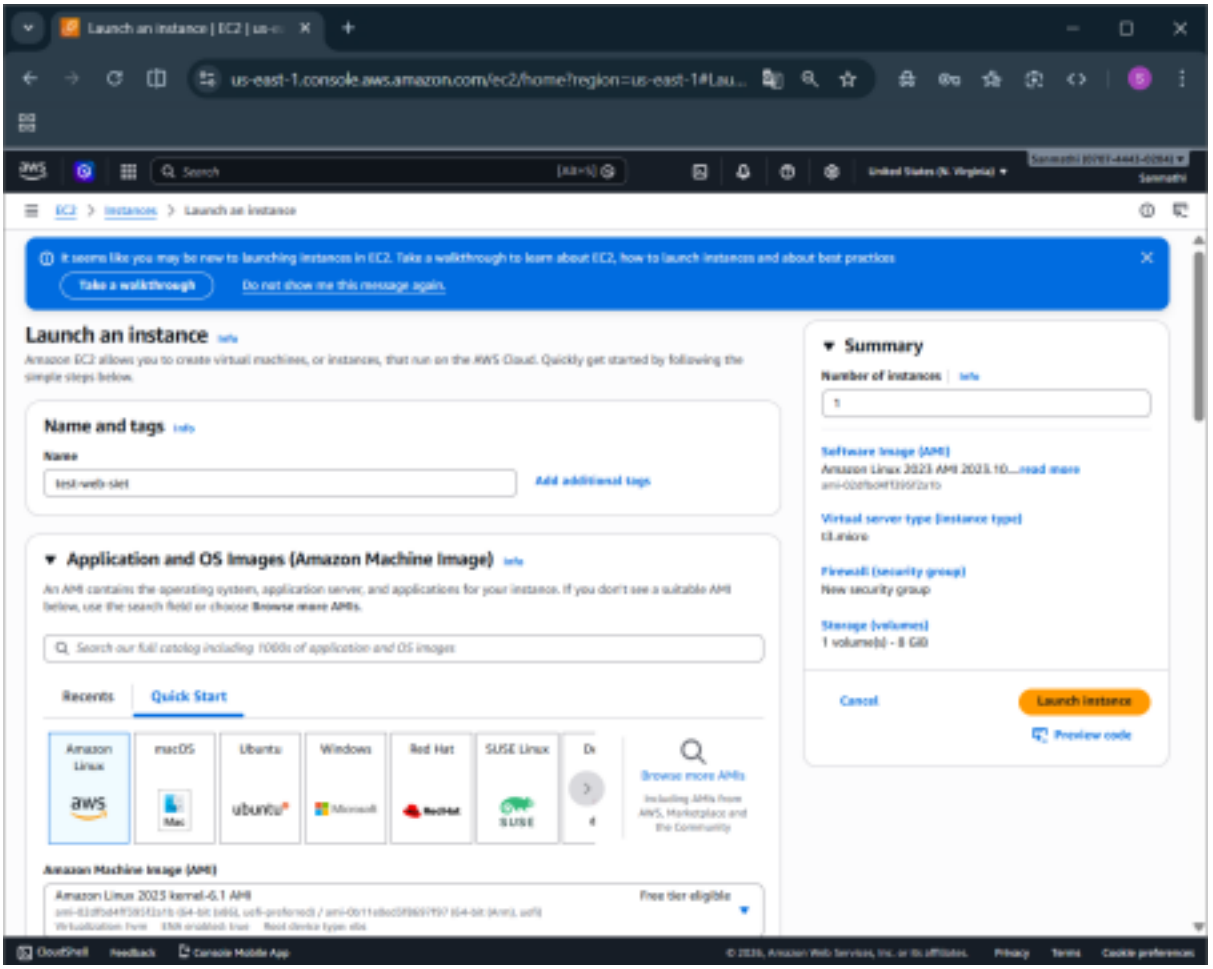

VPC | us-east-1
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-eas...
VPC > Route tables
VPC dashboard
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started
Endpoints
Endpoint services
Service networks
Route tables (1/3)
Actions
Create route table
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
test-siet_rt
rtb-0514ece3d82551963 / test-siet_rt
Details
Routes
Subnet associations
Edge associations
Route propagation
Tags
Explicit subnet associations (1)
Edit subnet associations
Name
Subnet ID
IPv4 CIDR
IPv6 CIDR
public-subnet
10.20.1.0/24
Subnets without explicit associations (1)
The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:
CloudShell
Feedback
Console Mobile App


Edit routes
Destination
Target
Status
Propagated
Route origin
Add route
Cancel
Preview
Save changes


Launch an instance | EC2 | us-e
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Lau...
Search
United States (N. Virginia)
Sanvathi
EC2 > Instances > Launch an instance
It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices
Take a walkthrough
Do not show me this message again.
Launch an instance Info
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Name
test-web-slet
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Amazon Linux 2023 kernel-6.1 AMI
Free tier eligible
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.10....read more
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Feedback
Console Mobile App
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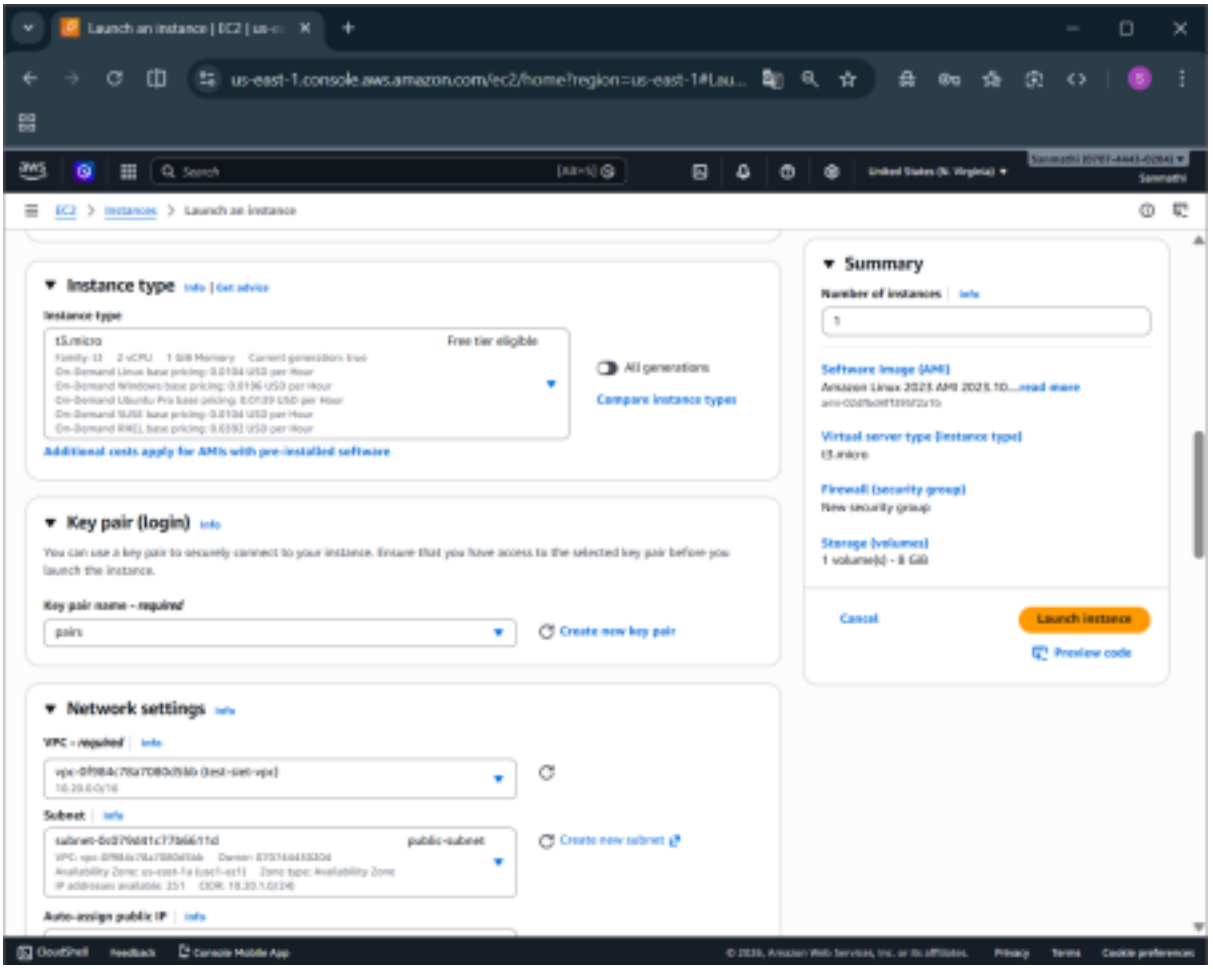
Launch an instance | EC2 | us-e
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Lau...
Search
United States (N. Virginia)
Samrathi
EC2 > Instances > Launch an instance
Instance type Info | Get advice
Instance type
t3.micro
Free tier eligible
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0104 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand RHEL base pricing: 0.0392 USD per Hour
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
pairs
Create new key pair
Network settings Info
VPC - required Info
vpc-0f984c76a708b0f6b (test-siet-vpc)
10.20.0.0/16
Subnet Info
subnet-0c079dd1c77b6611d
public-subnet
VPC: vpc-0f984c76a708b0f6b Owner: 070746433304
Availability Zone: us-east-1a (use1-az1) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.20.1.0/24
Create new subnet
Auto-assign public IP Info
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.10....read more
ami-02d5bd4f1399f2a1b
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
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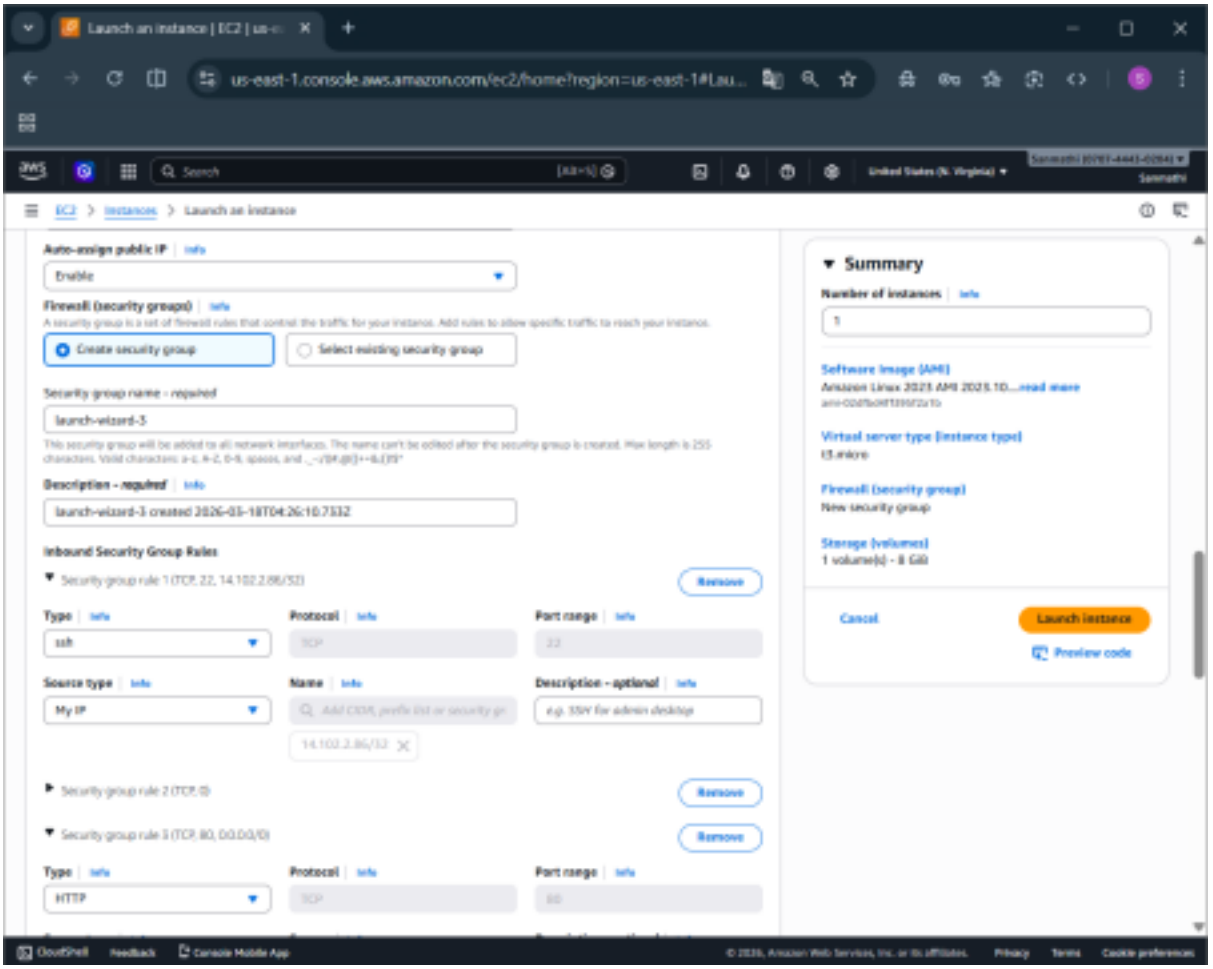
Launch an instance | EC2 | us-e
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Lau...
Search
United States (N. Virginia)
EC2 > Instances > Launch an instance
Auto-assign public IP | Info
Enable
Firewall (security groups) | Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Security group name - required
launch-wizard-3
This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-/:#,@[]+=&;{}!$*
Description - required | Info
launch-wizard-3 created 2026-03-18T04:26:10.733Z
Inbound Security Group Rules
Security group rule 1 (TCP, 22, 14.102.2.86/32)
Remove
Type | Info
ssh
Protocol | Info
TCP
Port range | Info
22
Source type | Info
My IP
Name | Info
Add CIDR, prefix list or security gr
Description - optional | Info
e.g. SSH for admin desktop
14.102.2.86/32
Security group rule 2 (TCP, 0)
Remove
Security group rule 3 (TCP, 80, 0.0.0.0/0)
Remove
Type | Info
HTTP
Protocol | Info
TCP
Port range | Info
80
Summary
Number of instances | Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.10....read more
ami-02d5bd4f1399f2a1b
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
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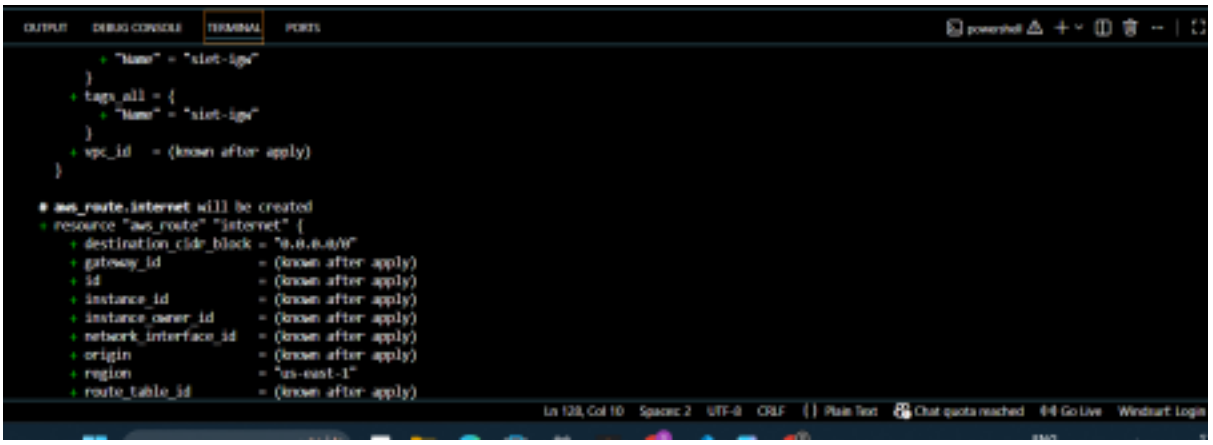
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
powershell
+ "Name" = "sirt-igw"
}
+ tags_all = {
+ "Name" = "sirt-igw"
}
+ vpc_id = (known after apply)
}
# aws_route.internet will be created
+ resource "aws_route" "internet" {
+ destination_cidr_block = "0.0.0.0/0"
+ gateway_id = (known after apply)
+ id = (known after apply)
+ instance_id = (known after apply)
+ instance_owner_id = (known after apply)
+ network_interface_id = (known after apply)
+ origin = (known after apply)
+ region = "us-east-1"
+ route_table_id = (known after apply)
Ln 128, Col 10
Spaces: 2
UTF-8
CRLF
Plain Text
Chat quota reached
Go Live
Windsurf: Login

```
aws_vpc.siet-vpc-1: Still creating... [00m10s elapsed
]
aws_vpc.siet-vpc-1: Creation complete after 12s [id=v
pc-0a712b09a5d42ed0b]
aws_internet_gateway.igw: Creating...
aws_route_table.public_rt: Creating...
aws_subnet.private_subnet: Creating...
aws_subnet.public_subnet: Creating...
aws_security_group.web_sg: Creating...
aws_route_table.public_rt: Creation complete after 4s
 [id=rtb-07849c4db94081391]
aws_subnet.private_subnet: Creation complete after 5s
 [id=subnet-01312d053287097e8]
aws_internet_gateway.igw: Creation complete after 6s
[id=igw-02266ecb862cc72bb]
aws_route.internet: Creating...
aws_subnet.public_subnet: Still creating... [00m10s e
lapsed]
aws_security_group.web_sg: Still creating... [00m10s
```

```
OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

  the verbosity of these warnings, use the -compact-warnings option.


Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_instance.siet-web-ec2: Creating...
aws_instance.siet-web-ec2: Still creating... [00m10s elapsed]
aws_instance.siet-web-ec2: Creation complete after 17s [id=i-0d2b574c1ae85f76c]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

siet-web-ec2_public_ip = "107.23.231.245"
PS D:\terra>
```

CODE:

VPC:

```
resource "aws_vpc" "main" {
  cidr_block       = "10.0.0.0/16"
  instance_tenancy = "default"

  tags = {
    Name = "main"
  }
```

```
}
```

SUBNET:

```
#Subnet - Public
resource "aws_subnet" "siet_public_subnet" {
  vpc_id     = aws_vpc.siet-vpc-1.id
  cidr_block = "10.0.1.0/24"
  availability_zone = "us-east-1a"

  tags = {
    Name = "siet_public_subnet"
    team = var.tag_team
    platform = var.tag_platform
  }
}

#Subnet - Private
resource "aws_subnet" "siet_private_subnet" {
  vpc_id     = aws_vpc.siet-vpc-1.id
  cidr_block = "10.0.2.0/24"
  availability_zone = "us-east-1b"

  tags = {
    Name = "siet_private_subnet"
    team = var.tag_team
    platform = var.tag_platform
  }
}
```

INTERNET GATEWAY:

```
#IGW
resource "aws_internet_gateway" "siet-igw" {
  vpc_id = aws_vpc.siet-vpc-1.id
```

```
  tags = {
    Name = "siet-igw"
    team = var.tag_team
    platform = var.tag_platform
  }
}
```

EC2:

```
resource "aws_instance" "siet-web-ec2" {
  ami           = var.my-ami
  instance_type = var.my-instance
  associate_public_ip_address = "true"
  key_name = "test-Keypair"
  subnet_id = aws_subnet.siet_public_subnet.id
  vpc_security_group_ids = [aws_security_group.siet-web-sg.id]

  tags = {
    Name = "siet-web-ec2"
    team = var.tag_team
    platform = var.tag_platform
  }
}


#security group
resource "aws_security_group" "siet-web-sg" {
  name        = "siet-web-sg"
  description = "Allow TLS inbound traffic and all outbound traffic"
  vpc_id      = aws_vpc.siet-vpc-1.id

  tags = {
    Name = "siet-web-sg"
    team = var.tag_team
    platform = var.tag_platform
  }
}
```

```
resource "aws_vpc_security_group_ingress_rule" "inbound_ssh" {
  security_group_id = aws_security_group.siet-web-sg.id
  cidr_ipv4         = "0.0.0.0/0"
  from_port         = 22
  ip_protocol       = "tcp"
  to_port           = 22
}
resource "aws_vpc_security_group_ingress_rule" "inbound_http" {
  security_group_id = aws_security_group.siet-web-sg.id
  cidr_ipv4         = "0.0.0.0/0"
  from_port         = 80
  ip_protocol       = "tcp"
  to_port           = 80
}

resource "aws_vpc_security_group_egress_rule" "allow_all_traffic_ipv4" {
  security_group_id = aws_security_group.siet-web-sg.id
  cidr_ipv4         = "0.0.0.0/0"
  ip_protocol       = "-1" # semantically equivalent to all ports
}

output "siet-web-ec2_public_ip" {
  value = aws_instance.siet-web-ec2.public_ip

}


#DB server
resource "aws_instance" "siet-db-ec2" {
  ami           = var.my-ami
  instance_type = var.my-instance
  key_name = "test-Keypair"
  subnet_id = aws_subnet.siet_private_subnet.id
  vpc_security_group_ids = [aws_security_group.siet-web-sg.id]

  tags = {
```

```
    Name = "siet-db-ec2"
    team = var.tag_team
    platform = var.tag_platform
  }
}
```

ROUTE TABLE:

```
#route table
resource "aws_route_table" "siet-route-table" {
  vpc_id = aws_vpc.siet-vpc-1.id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.siet-igw.id
  }

  tags = {
    Name = "siet-route-table"
    team = var.tag_team
    platform = var.tag_platform
  }
}
```

SUBNET ASSOCIATION:

```
#subnet association
resource "aws_route_table_association" "siet-public-subnet-association" {
  subnet_id      = aws_subnet.siet_public_subnet.id
  route_table_id = aws_route_table.siet-route-table.id
}
```

TERRAFORM AWS REGISTRY LINK:

https://registry.terraform.io/modules/terraform-aws-modules/vpc/aws/latest

```
module "vpc" {
  source = "terraform-aws-modules/vpc/aws"

  name = "test-vpc"
  cidr = "10.0.0.0/16"

  azs             = ["us-east-1a", "us-east-1b"]
  private_subnets = ["10.0.1.0/24"]
  public_subnets  = ["10.0.2.0/24"]

  tags = {
    Name = "test-vpc"
    team = var.tag_team
    platform = var.tag_platform
  }
}
```